

Further Integration Techniques

ANSWER KEY



Integration by substitution

- 1 Evaluate $\int 2x(x^2 + 1)^4 dx$ using the substitution $u = x^2 + 1$.
 $du = 2x dx$. So $\int (x^2 + 1)^4 \cdot 2x dx = \int u^4 du = u^5/5 + C = (x^2 + 1)^5/5 + C$.
- 2 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x dx$ using the substitution $u = \sin x$.
 $du = \cos x dx$. When $x = 0$, $u = 0$; when $x = \pi/2$, $u = 1$. $\int_0^1 u^3 du = [u^4/4]_0^1 = 1/4$.

Integration by parts

- 3 Evaluate $\int x e^x dx$ using integration by parts.
Let $u = x$, $dv = e^x dx$, so $du = dx$, $v = e^x$. $\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C = (x - 1)e^x + C$.
- 4 Evaluate $\int x \ln x dx$ using integration by parts.
Let $u = \ln x$, $dv = x dx$, so $du = (1/x) dx$, $v = x^2/2$.
 $\int x \ln x dx = x^2/2 \ln x - \int x^2/2 \cdot 1/x dx = x^2/2 \ln x - x^2/4 + C$.

Repeated integration by parts

- 5 Evaluate $\int x^2 e^x dx$, applying integration by parts twice.
First application: $u = x^2$, $dv = e^x dx$: $\int x^2 e^x dx = x^2 e^x - \int 2x e^x dx$. From the previous problem, $\int x e^x dx = (x - 1)e^x + C$, so $\int 2x e^x dx = 2(x - 1)e^x$. Result:
 $x^2 e^x - 2(x - 1)e^x + C = (x^2 - 2x + 2)e^x + C$.

Integration using trigonometric identities

- 6 Evaluate $\int \cos^2 x dx$ using the identity $\cos^2 x = (1 + \cos 2x)/2$.
 $\int (1 + \cos 2x)/2 dx = 1/2(x + \sin 2x/2) + C = x/2 + \sin 2x/4 + C$.

Integration using partial fractions

- 7 Evaluate $\int 1/(x^2 - 1) dx$ by first expressing the integrand in partial fractions.
 $1/(x^2 - 1) = 1/((x - 1)(x + 1)) = A/(x - 1) + B/(x + 1)$. Solving: $A = 1/2$, $B = -1/2$.
 $\int (1/2/(x - 1) - 1/2/(x + 1)) dx = 1/2 \ln|x - 1| - 1/2 \ln|x + 1| + C = 1/2 \ln|(x - 1)/(x + 1)| + C$.

Evaluating a definite integral

- 8 Evaluate $\int_1^e \ln x/x \, dx$, using the substitution $u = \ln x$.
 $du = (1/x) \, dx$. When $x = 1$, $u = 0$; when $x = e$, $u = 1$. $\int_0^1 u \, du = [u^2/2]_0^1 = 1/2$.
- 9 Evaluate $\int_0^\pi x \sin x \, dx$ using integration by parts.
Let $u = x$, $dv = \sin x \, dx$, so $du = dx$, $v = -\cos x$.
 $\int x \sin x \, dx = -x \cos x + \int \cos x \, dx = -x \cos x + \sin x$. Evaluating from 0 to π :
 $[-\pi \cos \pi + \sin \pi] - [0 + 0] = -\pi(-1) + 0 = \pi$.
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Integration involving inverse trigonometric results

- 10 Evaluate $\int 1/(1+x^2) \, dx$, and use it to evaluate $\int_0^1 1/(1+x^2) \, dx$.
 $\int 1/(1+x^2) \, dx = \arctan x + C$. So $\int_0^1 1/(1+x^2) \, dx = \arctan(1) - \arctan(0) = \pi/4 - 0 = \pi/4$.
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Integration by substitution with trigonometric functions

- 11 Evaluate $\int \sin^3 x \, dx$, using the identity $\sin^2 x = 1 - \cos^2 x$ and the substitution $u = \cos x$.
Rewrite: $\sin^3 x = \sin x(1 - \cos^2 x)$. Let $u = \cos x$, $du = -\sin x \, dx$.
 $\int \sin x(1 - \cos^2 x) \, dx = -\int (1 - u^2) \, du = -u + u^3/3 + C = -\cos x + \cos^3 x/3 + C$.
- 12 Evaluate $\int \tan x \, dx$, using the substitution $u = \cos x$.
 $\tan x = \sin x/(\cos x)$. Let $u = \cos x$, $du = -\sin x \, dx$.
 $\int \sin x/(\cos x) \, dx = -\int 1/u \, du = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.
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Integration by parts with a logarithmic integrand

- 13 Evaluate $\int_1^e \ln x \, dx$ using integration by parts.
Let $u = \ln x$, $dv = dx$, so $du = (1/x) \, dx$, $v = x$. $\int \ln x \, dx = x \ln x - \int x \cdot 1/x \, dx = x \ln x - x + C$.
Evaluating from 1 to e : $[e \ln e - e] - [1 \ln 1 - 1] = (e - e) - (0 - 1) = 0 + 1 = 1$.
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Integration using a Weierstrass-style substitution

- 14 Evaluate $\int 1/(5 + 4 \cos x) \, dx$ using the substitution $t = \tan(x/2)$, given that this substitution transforms $\cos x = (1 - t^2)/(1 + t^2)$ and $dx = 2/(1 + t^2) \, dt$, leading to the simplified integrand $2/(9 + t^2)$.
Substituting:
 $\int 1/(5 + 4(1 - t^2)/(1 + t^2)) \cdot 2/(1 + t^2) \, dt = \int 2/(5(1 + t^2) + 4(1 - t^2)) \, dt = \int 2/(9 + t^2) \, dt$. Using the standard result $\int 1/(a^2 + t^2) \, dt = (1/a) \arctan(t/a) + C$ with $a = 3$:
 $2 \cdot (1/3) \arctan(t/3) + C = (2/3) \arctan((1/3) \tan(x/2)) + C$.

Integration by parts with a trigonometric-exponential product

- 15** Evaluate $\int e^x \cos x \, dx$, applying integration by parts twice and solving the resulting equation for the original integral.
- Let $I = \int e^x \cos x \, dx$. First application: $u = \cos x$, $dv = e^x dx$, so $du = -\sin x \, dx$, $v = e^x$:
 $I = e^x \cos x + \int e^x \sin x \, dx$. Second application on $\int e^x \sin x \, dx$: $u = \sin x$, $dv = e^x dx$:
 $\int e^x \sin x \, dx = e^x \sin x - \int e^x \cos x \, dx = e^x \sin x - I$. Substituting back: $I = e^x \cos x + e^x \sin x - I$, so
 $2I = e^x(\cos x + \sin x)$, giving $I = e^x(\cos x + \sin x)/2 + C$.
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Finding the area between two curves using integration

- 16** Find the area of the region enclosed between $y = \sin x$ and $y = \cos x$ for $\pi/4 \leq x \leq 5\pi/4$ (the interval on which $\sin x \geq \cos x$).
- Area $= \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) \, dx = [-\cos x - \sin x]_{\pi/4}^{5\pi/4}$. At $x = 5\pi/4$:
 $-\cos 5\pi/4 - \sin 5\pi/4 = \sqrt{2}/2 + \sqrt{2}/2 = \sqrt{2}$. At $x = \pi/4$: $-\cos \pi/4 - \sin \pi/4 = -\sqrt{2}/2 - \sqrt{2}/2 = -\sqrt{2}$.
Area $= \sqrt{2} - (-\sqrt{2}) = 2\sqrt{2}$.
- 17** Evaluate $\int x^2 \ln x \, dx$ using integration by parts.
- Let $u = \ln x$, $dv = x^2 dx$, so $du = (1/x) dx$, $v = x^3/3$.
 $\int x^2 \ln x \, dx = x^3/3 \ln x - \int x^3/3 \cdot 1/x \, dx = x^3/3 \ln x - \int x^2/3 \, dx = x^3/3 \ln x - x^3/9 + C$.
- 18** A particle's velocity is given by $v(t) = t \cos t$ metres per second. Find the particle's displacement over the interval $0 \leq t \leq \pi$, using integration by parts.
- Displacement $= \int_0^\pi t \cos t \, dt$. Using integration by parts with $u = t$, $dv = \cos t \, dt$, so $du = dt$, $v = \sin t$:
 $\int t \cos t \, dt = t \sin t - \int \sin t \, dt = t \sin t + \cos t + C$. Evaluating from 0 to π :
 $[\pi \sin \pi + \cos \pi] - [0 \sin 0 + \cos 0] = [0 - 1] - [0 + 1] = -1 - 1 = -2$. Displacement $= -2$ metres.
- 19** This question has three parts. (a) Use the substitution $u = x^2 + 4$ to find $\int x/(x^2 + 4) \, dx$. (b) Use integration by parts to find $\int \arctan x \, dx$. (c) Hence, using the result of part (b) and the standard result $\int 1/(1+x^2) dx = \arctan x + C$, evaluate $\int_0^1 \arctan x \, dx$, giving an exact answer in terms of π and $\ln$.
- (a) $u = x^2 + 4$, $du = 2x \, dx$: $\int x/(x^2 + 4) \, dx = 1/2 \int 1/u \, du = 1/2 \ln|x^2 + 4| + C$. (b) Let $u = \arctan x$, $dv = dx$, so $du = (1/(1+x^2)) dx$, $v = x$:
 $\int \arctan x \, dx = x \arctan x - \int x/(1+x^2) \, dx = x \arctan x - 1/2 \ln(1+x^2) + C$ (using the result from part (a), with 4 replaced by 1). (c) Evaluating from 0 to 1:
 $[1 \cdot \arctan 1 - 1/2 \ln 2] - [0 - 1/2 \ln 1] = \pi/4 - 1/2 \ln 2$.
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Reduction formulas

- 20 Given the reduction formula $I_n = \int x^n e^x dx = x^n e^x - n I_{n-1}$ (obtained via integration by parts), and that $I_0 = e^x + C$, use it to find $I_2 = \int x^2 e^x dx$.
- $I_1 = x^1 e^x - 1 \cdot I_0 = x e^x - e^x$ (ignoring the constant until the final step).
- $I_2 = x^2 e^x - 2 I_1 = x^2 e^x - 2(x e^x - e^x) = x^2 e^x - 2x e^x + 2e^x + C = (x^2 - 2x + 2)e^x + C$, matching the result found earlier in this worksheet by direct repeated integration by parts.
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Integration by substitution with a linear inner function

- 21 Evaluate $\int (3x + 2)^5 dx$ using the substitution $u = 3x + 2$.
- $du = 3 dx$, so $dx = du/3$. $\int u^5 \cdot (1/3) du = (1/3)(u^6/6) + C = u^6/18 + C = (3x + 2)^6/18 + C$.
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Integration using the sine double-angle identity

- 22 Evaluate $\int \sin^2 x dx$ using the identity $\sin^2 x = (1 - \cos 2x)/2$.
- $\int (1 - \cos 2x)/2 dx = 1/2(x - \sin 2x/2) + C = x/2 - \sin 2x/4 + C$.
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Volume of revolution using integration

- 23 The region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is rotated 360° about the x-axis. Find the volume of the resulting solid, using $V = \pi \int_a^b y^2 dx$.
- Since $y = \sqrt{x}$, $y^2 = x$. $V = \pi \int_0^4 x dx = \pi [x^2/2]_0^4 = \pi(8 - 0) = 8\pi$ cubic units.
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Substitution with a shifting definite integral bound

- 24 Evaluate $\int_0^1 x \sqrt{x^2 + 1} dx$ using the substitution $u = x^2 + 1$.
- $du = 2x dx$, so $x dx = du/2$. When $x = 0$, $u = 1$; when $x = 1$, $u = 2$. So the integral becomes $\int \sqrt{u} \cdot (1/2) du$, with u running from 1 to 2. Since $\int \sqrt{u} du = (2/3)u^{3/2} + C$, this gives $(1/2)(2/3)(2^{3/2} - 1) = (1/3)(2\sqrt{2} - 1)$.
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Choosing an appropriate integration technique

25 Which integration technique is most appropriate for evaluating $\int x^3 \sin x \, dx$?

- a. Integration by parts, applied three times (reducing the power of x each time) ✓
- b. Substitution $u = \sin x$
- c. Partial fractions
- d. The double-angle identity for $\sin$

Since the integrand is a product of a polynomial and a trigonometric function, integration by parts is appropriate — and because the polynomial factor is x^3 (not x^1), the process must be repeated three times, each time reducing the power of x by one, until the polynomial factor disappears.